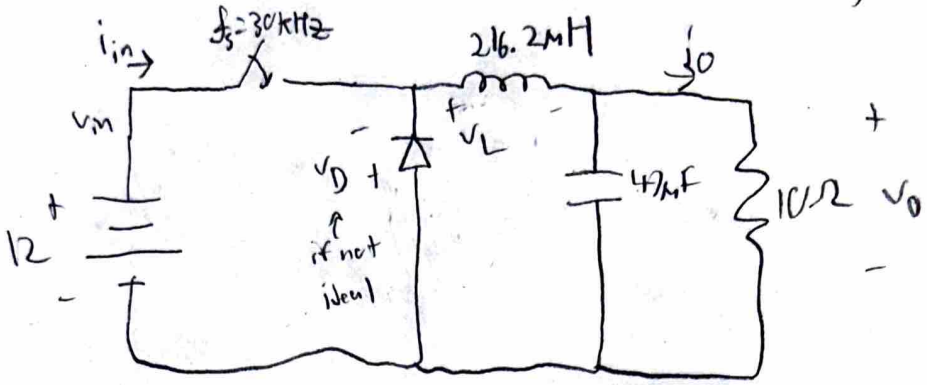
Prelab 6 Dylan Thomas

a.



b.

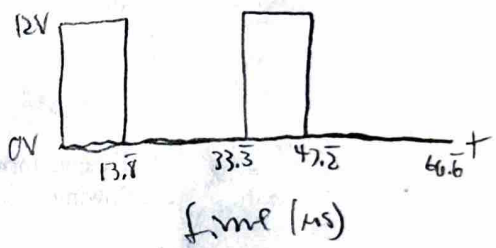
$V_o = 5V$ $V_o = V_m(D) \rightarrow 5 = 12D \rightarrow D = \frac{5}{12} \rightarrow T_{on} = D\frac{1}{f_s} = \boxed{13.889\,\mu s}$

$V_m = 12V$

c.

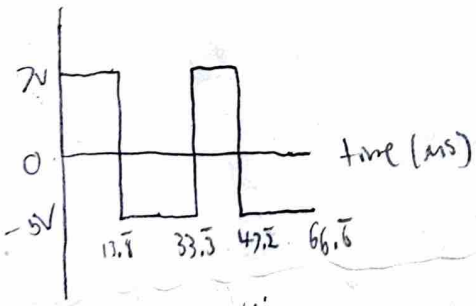
i.

ii.



iii.

12 − 5 = 7V across inductor



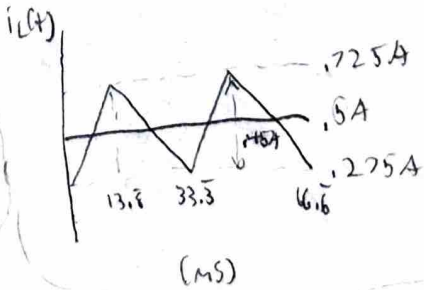
iv.

$I_{LON} = \frac{(12-5)(\frac{5}{12}T)}{216.2\mu H} = .450A$

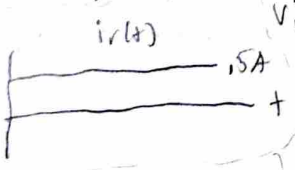
$I_{LOFF} = \frac{-5(1-\frac{5}{12})T}{216.2\mu H} = -.450A$

$I_L = I_o = \frac{5}{10} = .5A$ $.5 + \frac{.45}{2} = .725A$

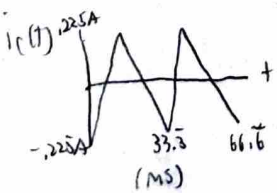
$.5 - \frac{.45}{2} = .275A$



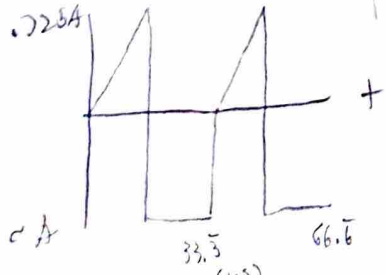
v.



vi.



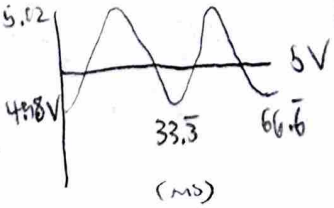
vii.



viii.

$\Delta V_o = \frac{\Delta i_L}{8Cf_s}$

$= \frac{.45A}{8(47\mu)(30k)} = .040V$



D. $I_{crit} = \frac{.45}{2} = .225A$

$R_{crit} = \frac{5}{.225} = 22.2\Omega$

R_L of 22.2Ω or more will cause DCM